

Muhammad Taha Khan

Full Stack Developer — Generative AI Engineer

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Professional Summary

Full Stack AI Engineer with experience building production AI systems, RAG pipelines, conversational voice agents, CRM platforms and full-stack SaaS applications using Next.js, React, Node.js, Python, FastAPI, LangChain and LLMs.

Experience

Full Stack AI Engineer

Zentri AI

Dec 2025 – Present

London, UK | Remote

- Designed and built Zentri AI's production CRM platform, AI chatbot, and conversational voice agent as the founding engineer, leading a team of two developers to deliver AI solutions for UK healthcare and dental clinics.
- Architected real-time voice AI pipelines using Pipecat, Deepgram (STT/TTS), WebSockets, and LLM orchestration, enabling natural phone conversations while supporting a platform that has processed over **22,100+ customer enquiries**.
- Developed Retrieval-Augmented Generation (RAG) chatbots using LangChain, embeddings, vector search, and automated CRM synchronization, powering **38,480+ chatbot conversations** and contributing to more than **\$6.63M in tracked lead revenue**.
- Built scalable full-stack applications using Next.js, React, Node.js, Express, MongoDB, FastAPI, Docker, and AWS while improving deployment reliability and reducing manual operational workflows.

Full Stack Developer

TDMC

Jun 2025 – Nov 2025

London, UK | Remote

- Developed AI-powered SaaS web applications using Next.js, React, Node.js, Express, MongoDB, and REST APIs.
- Integrated generative AI features into business workflows, reducing manual lead handling and establishing the technical foundation that later evolved into Zentri AI.

Frontend Developer

SAHL, NASTP

Nov 2024 – Dec 2024

Remote

- Rebuilt the company's website using React with responsive layouts across mobile, tablet, and desktop devices, improving usability and cross-device consistency.
- Optimized frontend performance, resulting in faster page rendering and a smoother user experience.

MERN Developer Intern

WALEE Lab, SEECs

Jun 2024 – Aug 2024

Islamabad, Pakistan

- Developed responsive MERN applications using React, Node.js, Express, MongoDB, and REST APIs.
- Improved backend query efficiency and frontend asset loading, enhancing performance across multiple production features.

Projects

Zentri AI

AI CRM, Chatbot & Voice Agent Platform

Dec 2025 – Present

LLMs, RAG, Next.js, MERN

- Designed and developed a production AI platform combining CRM, conversational chatbot, and voice AI capabilities using Next.js, React, Node.js, MongoDB, FastAPI, and REST APIs.
- Engineered a low-latency voice AI pipeline using Pipecat, Deepgram (STT/TTS), WebSockets, and LLM orchestration for real-time phone conversations.
- Built a Retrieval-Augmented Generation (RAG) system using LangChain, vector embeddings, and Pinecone/FAISS while automating CRM synchronization through webhook workflows.

- Implemented backend architecture, authentication, booking automation, analytics dashboards, and AI-powered lead management.

Ball Wijzer – Football Analytics Chatbot

Final Year Project

Python, YOLO, ByteTrack, LangChain, LLMs

- Developed an AI-powered football analytics platform combining computer vision and conversational AI.
- Implemented player detection, tracking, team classification, and event recognition using YOLO, ByteTrack, OpenCV, and SoccerNet.
- Integrated LangChain with football APIs for live statistics, contextual search, and natural-language interaction.

Other Projects

- **Bank LLM Assistant** — Fine-tuned LLaMA 3.2 3B and implemented a LangChain + FAISS Retrieval-Augmented Generation pipeline using Streamlit.
- **Metro Shoes Clone** — Built a full-stack MERN e-commerce application with JWT authentication, payment integration, and order management.
- **Music Tutorial Platform** — Developed a Next.js learning platform with authentication, dashboards, and progress tracking.
- **Portfolio Website** — Designed and deployed a developer portfolio using Next.js, Tailwind CSS, Framer Motion, and Vercel.

Education

National University of Sciences and Technology (NUST)

Nov 2021 – Jun 2025

Bachelor of Engineering in Software Engineering

Islamabad, Pakistan

- Relevant Coursework: Large Language Models, Machine Learning, Deep Learning, Natural Language Processing, Object Oriented Programming, Data Structures, Algorithms, Database Systems, Software Engineering.

Abbottabad Public School

2019 – 2021

Intermediate (Pre-Engineering)

Abbottabad, Pakistan

Technical Skills

Programming Languages: JavaScript, TypeScript, Python, C++, HTML5, CSS3

Frontend: React.js, Next.js, Tailwind CSS, Responsive Design

Backend: Node.js, Express.js, FastAPI, REST APIs

Artificial Intelligence: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, Prompt Engineering, Fine-tuning (LLaMA), Conversational AI

Computer Vision: YOLO, ByteTrack, DeepSort, OpenCV

Voice AI: Pipecat, Deepgram, Speech-to-Text (STT), Text-to-Speech (TTS), Voice Agents

Databases & Vector Search: MongoDB, MySQL, Pinecone, FAISS

Cloud & DevOps: Docker, AWS, Git, GitHub, Vercel, Railway

Languages: English (IELTS 8.0), Urdu (Native)

Leadership & Extracurricular Activities

Founding Engineer, Zentri AI – Currently leading a team of two developers while driving architecture, product development, deployment, and AI engineering initiatives.

Organising Member, Palestine Relief Fair (NUST) – Coordinated fundraising activities, vendor management, and event logistics, contributing to the successful collection of PKR 150,000 for Gaza through the Al-Khidmat Foundation.

NUST Cricket Team Member – Represented the university in inter-university cricket tournaments while balancing academics and software engineering projects.